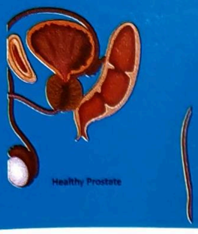
CHAPTER

11



Common Urinary Diseases

Learning Objectives

- ⇨ To learn about brief pathophysiology of urinary diseases
- ⇨ To learn about the causes, clinical features of urinary tract infection (UTI), prostatomegaly, nephrotic syndrome, nephritic syndrome, acute kidney injury, chronic kidney disease (CKD)
- ⇨ To learn the method of physical examination in above said diseases for the diagnosis
- ⇨ To make the interpretation of laboratory investigations in above said diseases

Introduction

Excretory system is one of the major system, primarily functions as the eliminator of the metabolites formed in the body. It has associated functions such as regulation of blood pressure of the body, maintaining the haemoglobin concentration etc. This system has kidney and ureter one on both sides, a urinary bladder and urethra. Each of the organ afflicted may produce typical symptoms. For example, affliction of the kidney may show increased frequency of micturition, haematuria; affliction of ureter may have pain which radiates from loin to groin; affliction of urinary bladder may be associated with pain in the lower abdomen, haematuria; affliction of urethra may be associated with burning micturition.

- Clinical symptoms of these disorders will depend upon the pathophysiology of renal injury and will often initially identified as a complex of symptoms, abnormal physical examination findings and laboratory changes that together make possible the identification of specific syndromes. These renal syndromes may arise as the consequence of a systemic illness or can occur as a primary renal disease.
- The duration and severity of the disease will affect these findings and typically include one or more of the following:
  - ➢ Disturbances in urine volume (oliguria, anuria or polyuria)
  - ➢ Abnormalities of urinary sediments (RBCs, WBCs, casts and crystals)
  - ➢ Abnormal excretion of serum proteins
  - ➢ Reduction in glomerular filtration rate (GFR) i.e. azotemia
  - ➢ Presence of hypertension and/ or expanded total body volume (oedema)
  - ➢ Electrolyte abnormalities
  - ➢ Fever/ pain in some of the occasions
- Carefully, above factors should be evaluated thoroughly to make the accurate diagnosis of renal disease.

Urinary Tract Infection

Urinary tract infections **(see Figure 20)** are the group of diseases where microorganisms afflict the urinary tract. This can start either from the kidney or through urethra. These are characterized by increased frequency of micturition, burning sensation during micturition, pain in the back or pain in the flank, pain in the supra pubic region, haematuria. Urinary tract infections can spread to other organs of the urinary tract i.e. from kidney to urethra or vice versa.

Causes:

- Septicemia
- Excess sexual intercourse with infected individuals
- Any conditions which is responsible for the retention of urine such as prostatomegaly or urethral stricture or occlusion of urinary calculus or cystocele in females

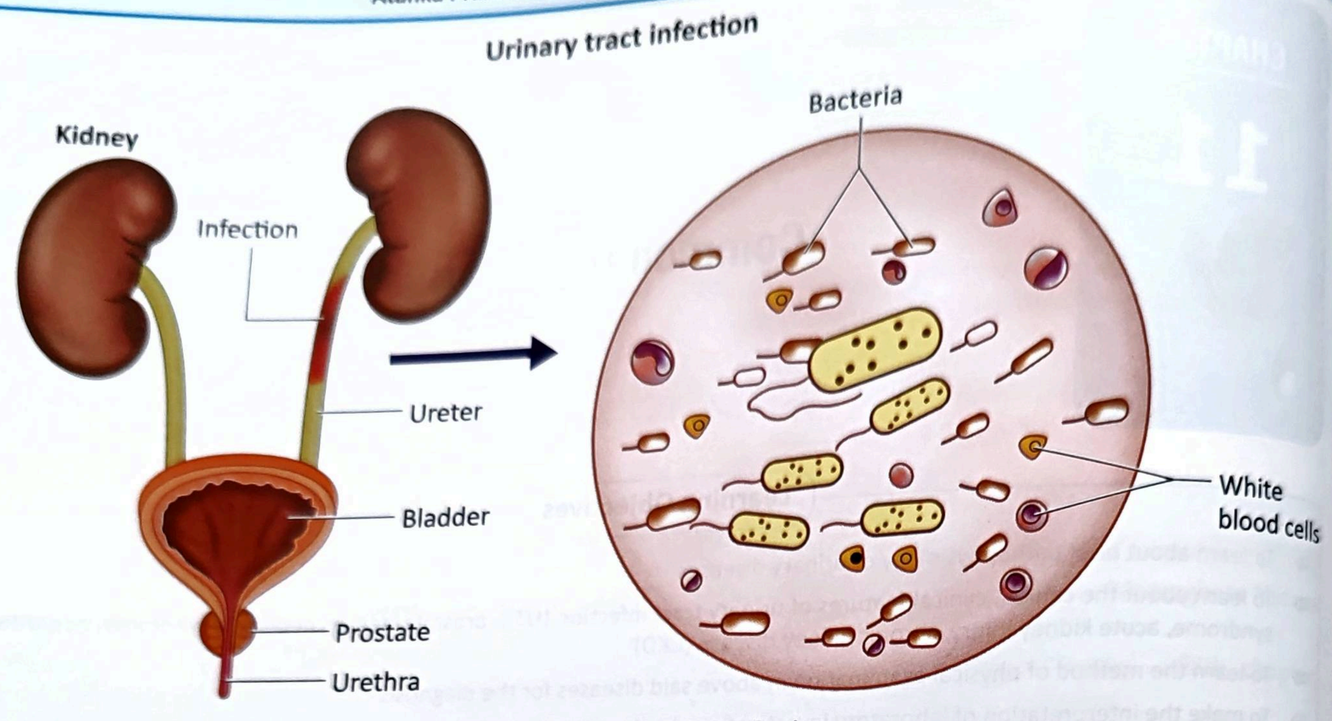


Figure 20 : Urinary Tract Infection

Clinical features:

- Depending upon the involvement of organ afflicted different clinical features can be seen. Increased frequency of micturition; whereas affliction of ureter, the patient will have pain from the flank which radiate from loin to groin; in case of affliction of urinary bladder, the patient will have pain in the supra pubic region; and affliction of urethra leads to the burning micturition.
- In case of urinary tract infection, the person will have one or two or all symptoms.

Physical examination:

- **Murphy's kidney punch (renal angle test):** In sitting posture, the patient folds his arms in front so that the back is stretched enough for better palpation. The clinician presses his thumb on the renal angle formed by the lower border 12th rib and outer border of erector spinae. This is of great value in determining tenderness of the kidney. This is usually indicating the calculi which is one of the cause of UTI.
- **Per abdominal examination:** Palpation of per abdominal examination is done on the right lumbar and left lumbar region to analyse the status of the ureter and palpation of the supra pubic region to analyse the status of the urinary bladder.
- Urethra can be examined to analyse the inflammation of the urethra or any ulcer of the urethra or the presence of any urethral anomalies such as hypospadias. In the females, the examination of external genital organs to exclude the cystocele.

Laboratory investigations:

- Complete blood count:
- Urine analysis: microscopic examination of urine particular
- Urine culture:
- Physical examination of urine:

Prostatomegaly

The term prostatomegaly **(see Figure 21)** refers to enlargement of prostate gland. Normally occurring prostatic enlargement is benign prostatic hyperplasia (BPH). This is non malignant adenomatous overgrowth of the periurethral gland. Diagnosis is done by digital examination primarily and based on symptoms; cystoscopy, transrectal ultrasonography, urodynamics and other studies. This is usually seen in men aged 55 to 74 without prostatic cancer. Aetiology is usually unknown.

Pathophysiology:

- Multiple fibroadenomatous nodules develop in the periurethral region of the prostate. This leads to the narrowing and lengthening of prostatic urethra. This leads to the hypertrophy of bladder detrusor (obstruction to the flow of urine), trabeculation, cellule formation and diverticula.
- This may lead to incomplete bladder emptying, thus resulting in formation of calculus and infection. Prolonged obstruction may lead to hydronephrosis and compromise in renal function.

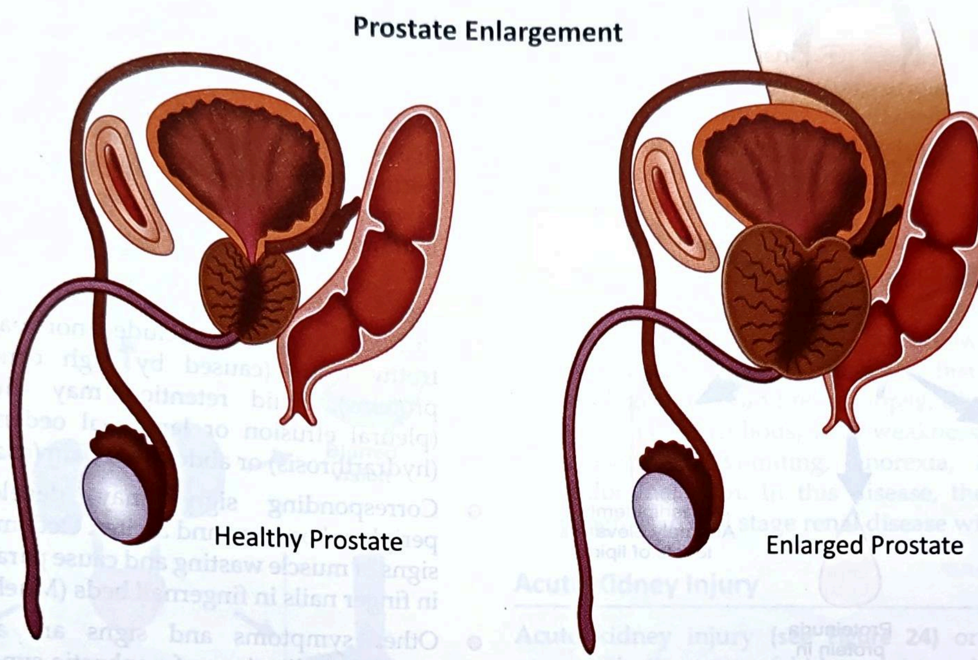


Figure 21 : Image of normal prostate and begign prostate enlargement

Clinical features:

- Symptoms include progressive urinary frequency, urgency, nocturia due to incomplete emptying and rapid refilling of the bladder. Pain and dysuria are not observed. Hesitancy and intermittency are usually observed secondary to decreased size and force of the urinary stream.
- Sensations of incomplete emptying, terminal dribbling, overflow incontinence or complete urinary retention may occur.
- Straining to void can cause congestion of the superficial veins of the prostatic urethra and trigone, which may rupture and cause haematuria. Straining may also cause vasovagal syncope in acute condition.
- Some patients may present with sudden, complete urinary retention, with marked abdominal discomfort and bladder retention. This may be associated with prolonged attempts to retain urine, immobilization, exposure to cold etc.

Diagnosis is done by history, digital rectal examination, urinalysis and culture, prostate specific antigen level , bladder ultrasonography, transrectal biopsy .

Nephrotic Syndrome

Nephrotic syndrome **(see Figure 22)** is urinary excretion of > 3gm of protein/ day due to a glomerular disorder. It is more common among children and has both primary and secondary causes. Diagnosis is by determination of urine protein/ creatinine ratio in a random urine sample or measurement of urinary protein in 24 hour urine collection; cause is diagnosed based on history, physical examination, serologic testing and renal biopsy. Prognosis may vary depending upon the cause.

Etiology of nephrotic syndrome:

- This disease can occur at any age, but more commonly observed in children, mostly between the age of 1 and ½ years to 4 years. At younger ages, boys are affected more frequently than girls; but both genders are afflicted with this disease equally when grew older.
- The following are primary diseases of nephrotic syndrome:
  - Minimal change disease
  - Focal segmental glomerulosclerosis
  - Membranous nephropathy
- Secondary causes are
  - **Metabolic:**
- Amyloidosis
- Diabetes mellitus
  - **Immunological:**
- Systemic lupus erythematosus
- Polyarthritis nodosa
- Serum sickness
- Sjogren's syndrome
- Erythema multiforme
  - **Idiopathic:**
- Sarcoidosis
  - **Neoplastic:**

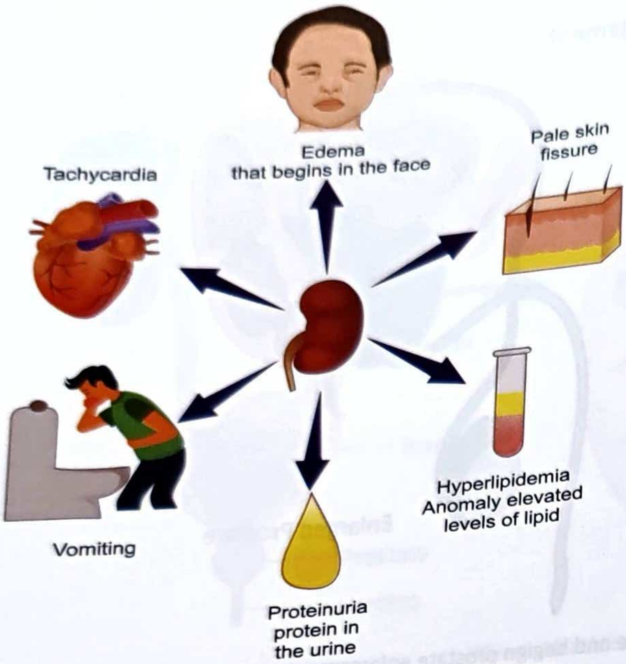


Figure 22: Clinical features of nephrotic syndrome

- Leukemia
- Lymphoma
- Melanoma
- Multiple myeloma

➤ ***Drug related:***

- Gold
- Heroin
- Lithium
- Mercury
- NSAIDs (non steroidal anti inflammatory drugs)

➤ ***Bacterial:***

- Infective endocarditis
- Leprosy
- Syphilis

➤ ***Protozoa:***

- Filariasis
- Malaria
- Helminths infestation

➤ ***Viral:***

- Hepatitis B and C
- Herpes zoster
- HIV infection

➤ ***Diabetic nephropathy***

➤ ***Preeclampsia***

Pathophysiology:

- Proteinuria occurs due to changes in the capillary endothelial cells of glomerular basement membrane.
- The mechanism of damage to the endothelial cells is unknown in primary and secondary glomerular diseases. But T cells may up-regulate a circulating permeability factor or down-regulat an inhibitor of permeability factor in response to unidentified immunogens and cytokines.

Clinical features:

- Primary symptoms include anorexia, malaise and frothy urine (caused by high concentrations of proteins). Fluid retention may cause dyspnoea (pleural effusion or laryngeal oedema), arthralgia (hydrarthrosis) or abdominal pain (ascites).
- Corresponding signs may develop including peripheral oedema and ascites. Oedema may obscure signs of muscle wasting and cause parallel white lines in finger nails in fingernail beds (Muehrcke's lines).
- Other symptoms and signs are attributable to many complications of nephrotic syndrome such as oedema, infection such as cellulitis, anemia, protein malnutrition or chronic kidney disease.

Diagnosis of the disease: Diagnosis is suspected in patients with oedema and proteinuria on urinalysis and confirmed by proteinuria test and 24 hours measurement of proteins. This may be seen in SLE, preeclampsia, cancer. When the cause is unclear, then additional serological testing and proteinuria test is performed to confirm the diagnosis.

- Proteinuria ≥ 3 gm/ 24 hours
- Serological testing and renal biopsy until the cause is clinically obvious.

➤ **Elevated HbA1C level in case of diabetes mellitus;** ANA tests; hepatitis B and C tests may help in the assessment of secondary causes of nephrotic syndrome.

➤ **Renal biopsy:** is performed in the diagnosis of chronic kidney disease.

Prognosis:

- Prognosis vary according to the cause. Complete remissions may occur spontaneously or with treatment. The prognosis is generally favourable in corticosteroid responsive disorders.
- Prognosis is worsened by infection; hypertension; significant azotemia; hematuria; cerebral thrombosis; pulmonary thrombosis; peripheral thrombosis; renal vein thrombosis.

Nephritic Syndrome

Nephritic syndrome **(see Figure 23)** is defined by haematuria and the presence of dysmorphic RBCs and RBC casts on microscopic examination of urinary sediment. Often one or more of the following elements are present: mild to moderate proteinuria, oedema, hypertension, elevated serum creatinine and oliguria. Diagnosis is done by history, physical examination and sometimes by renal biopsy.

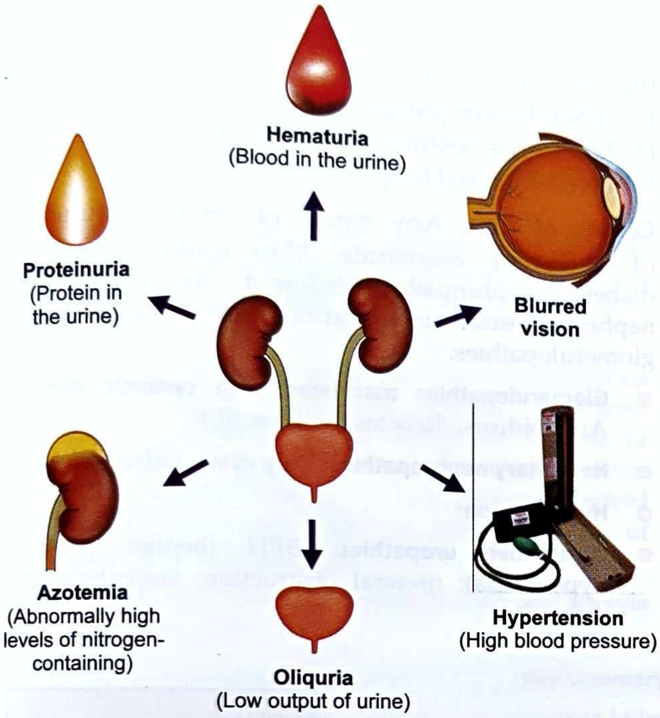


Figure 23 : Clinical features of nephritic syndrome

- **Nephritic syndrome is a manifestation of glomerular inflammation** i.e. glomerulonephritis (GN) and occurs at any age. Causes of the disease may be either primary (IgA glomerulonephropathy, fibrillary GN etc or secondary such as infective (bacterial, viral or parasitic), connective tissue disorders such as SLE etc.

Clinical features of the disease

- Most common manifestation is persistent or recurrent macroscopic haematuria or asymptomatic microscopic haematuria with mild proteinuria. Gross haematuria begins in 1 to 2 days after a febrile mucosal (upper respiratory, sinus, enteral) illness, thus mimicking postinfective GN. This is seen in IgA nephropathy. Diagnosis is by clinical findings and urinalysis and sometimes renal biopsy. IgA nephropathy progresses slowly, renal insufficiency and hypertension develop within 10 years in 15 to 20% of the patients; progression to end stage renal disease occurs in 20% of the patients after 20 years of the initial onset.
- Postinfectious glomerulonephritis (PIGN) occurs in children between 5 to 15 years. Symptoms range from asymptomatic haematuria and mild proteinuria to full blown nephritis with microscopic or gross haematuria (cola coloured, brown, smoky or frank blood in the urine), proteinuria, oliguria, oedema, hypertension and renal insufficiency. Diagnosis is done by clinical evidence of recent infection; urinalysis showing GN. Normal renal function is retained or regained in 85 to 95% of the patients. GFR returns to normal over 1 to 3 months; but proteinuria may persist for 6 to 12 months and microscopic haematuria may persist for several years.
- Rapidly progressive glomerulonephritis (RPGN) causes microscopic glomerular crescent formation with progression to renal failure within weeks to months. Diagnosis is based on history, urinalysis, serological tests and renal biopsy. Clinical symptoms are usually insidious, with weakness, fatigue, fever, nausea and vomiting, anorexia, arthralgia and abdominal pain. In this disease, the patients may progress to end stage renal disease within 6 months.

Acute Kidney Injury

Acute kidney injury **(see Figure 24)** or acute tubular necrosis is the type of kidney injury characterized by acute tubular cell injury and dysfunction. Common causes are hypotension causing renal hypoperfusion and nephrotoxic drugs. The condition is usually asymptomatic unless it causes renal failure. The diagnosis is suspected when azotemia develops after a hypotensive event, severe sepsis or drug exposure and is distinguished from pre renal azotemia by laboratory testing and response to volume expansion.

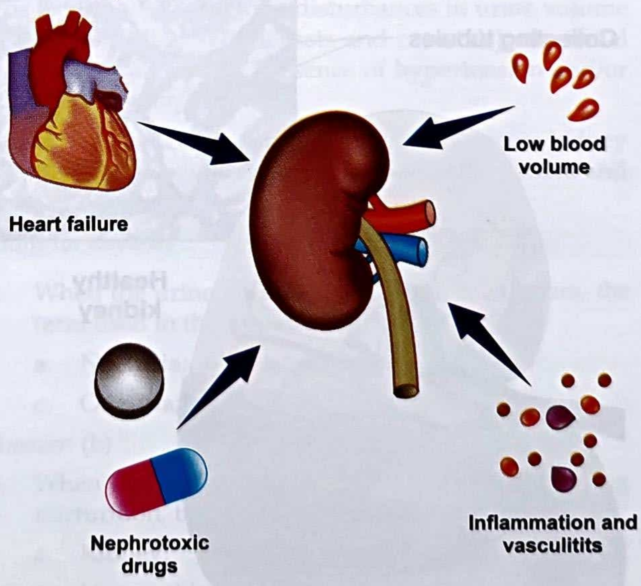


Figure 24 : Causes and features of acute kidney injury

Causes:

- Hypotension (ischemic ATN)
- Nephrotoxins
- Sepsis

- Major surgery
- Third degree burns
- Disorders resulting in other endogenous toxins such as tumour lysis or multiple myeloma

Pathogenesis

- Massive volume loss as in case of hypovolemic shock or hemorrhagic shock or pancreatitis
- Serious surgeries and advanced hepatobiliary diseases, poor perfusion states, advanced age who is at risk for aminoglycoside toxicity.

Clinical features

Asymptomatic initially but may cause symptoms or signs of acute renal failure, typically oliguria (urine volume <500 ml in 24 hours)

Diagnosis: differentiation from prerenal azotemia, based on laboratory findings mainly, and in the case of blood or fluid loss, response to volume expansion excludes the diagnosis acute kidney injury.

Chronic Kidney Disease

This is also being termed as chronic renal failure or CKD. **(see Figure 25)** This is a long standing, progressive deterioration of renal function. Symptoms develop slowly and include anorexia, nausea, vomiting, stomatitis, dysgeusia, nocturia, lassitude, fatigue, pruritus, decreased mental acuity, muscle twitches and muscle cramps, water retention, undernutrition, GI ulceration and bleeding; peripheral neuropathies, seizures. Diagnosis is done by laboratory testing of renal function, sometimes followed by renal biopsy.

Causes of CKD: Any cause of renal dysfunction of sufficient magnitude. Most common cause is diabetic nephropathy, followed by hypertensive nephroangiosclerosis and various primary and secondary glomerulopathies.

- **Glomerulopathies associated with systemic disease:** Amyloidosis; diabetes mellitus; SLE
- **Hereditary nephropathies:** Poly cystic kidney diseases
- **Hypertension:**
- **Obstructive uropathies:** BPH (benign prostatic hyperplasia); ureteral obstruction; vesicoureteral reflux

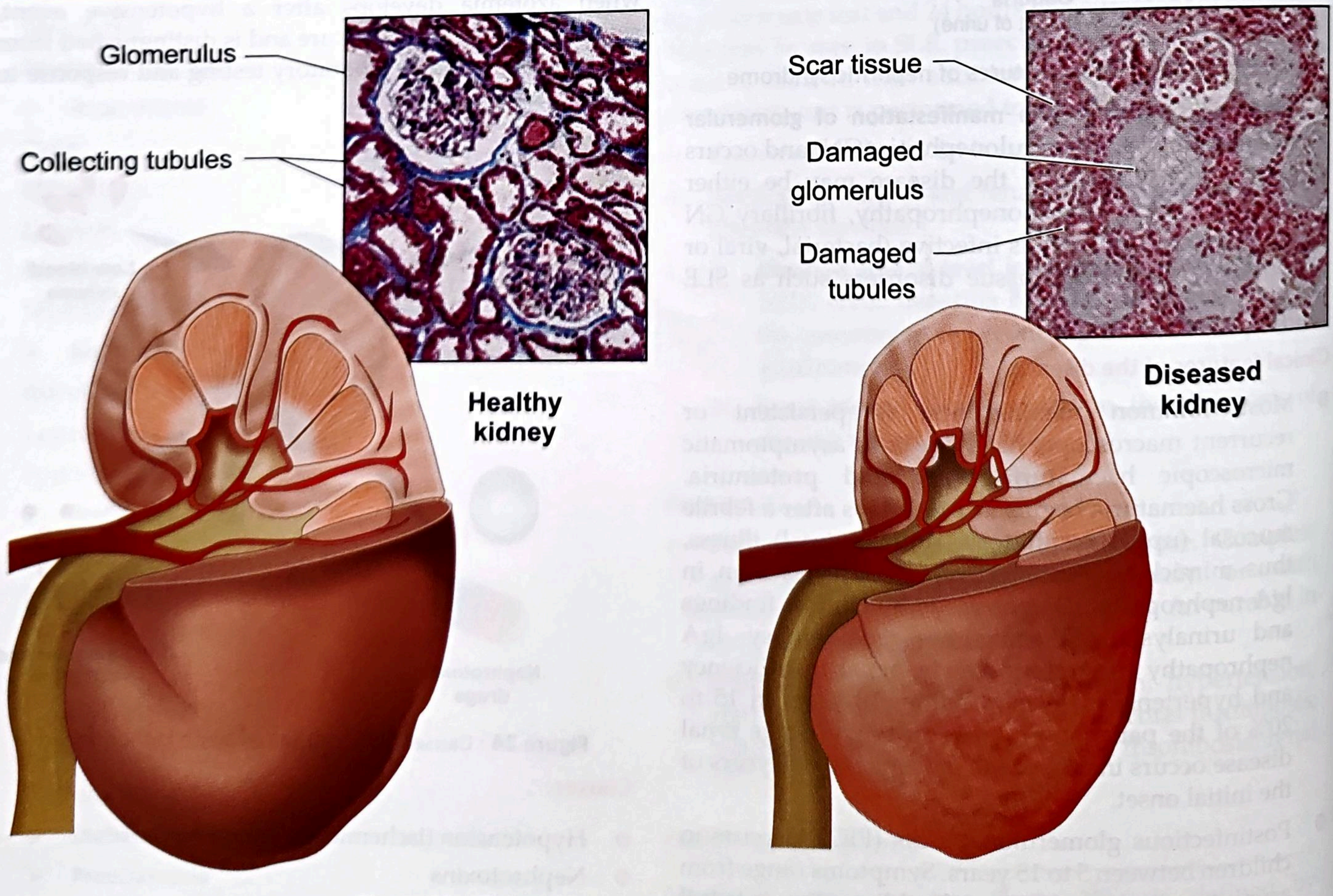


Figure 25 : Clinical features of chronic kidney disease (CKD)

Pathogenesis: CKD can be categorized into diminished renal reserve; renal insufficiency or renal failure (end stage renal disease)

- Initially, renal tissue loses function, there are few abnormalities because of remaining tissues taking the functions of the diseased nephrons. 75% of the renal tissue damage leads to the a fall of GFR to only 50% of the normal.
- Decreased renal function interferes with the kidney's ability to maintain fluid and electrolyte homeostasis. Plasma concentrations of creatinine and urea are going to rise on the reduction of GFR.
- Despite a diminishing GFR, sodium ions and water balance is well maintained by increased fractional excretion of sodium ions and a normal response to thirst.
- Renal osteodystrophy (abnormal bone mineralization resulting from hyperparathyroidism, calcitriol deficiency, elevated serum phosphate or low or normal serum calcium level usually takes the form of increased bone turn over due to hyperparathyroid bone disease. There may be the possibility of osteopenia or osteomalacia.
- Moderate acidosis (excess carbonic acid) and anemia may be observed.

Clinical features

- Patients with mildly diminished renal reserve are asymptomatic. Even the patients with mild to moderate renal insufficiency may have no symptoms despite elevated BUN and creatinine. Nocturia is often noted due to failure to concentrate urine.
- With more severe renal insufficiency, neuromuscular symptoms may be present, including coarse muscular twitches, peripheral sensory and motor neuropathies, muscle cramps, hyperreflexia and seizures may be observed. Anorexia, nausea, vomiting, weight loss, stomatitis are almost uniformly present. Pruritus may be uncomfortable.
- In advanced CKD, pericarditis, GI ulceration and bleeding are common.
- Diagnosis is done by assessment of serum electrolytes, creatinine, CBC, urinalysis, ultrasonography and sometimes renal biopsy.
- Prognosis is predicted in most cases by the degree of proteinuria. Patients with nephrotic range proteinuria (≥ 3g/ 24 hours) usually have poorer prognosis and progress to renal failure rapidly. Patients with proteinuria (≤ 1.5 g/ 24 hours) progression occurs very slowly.

Key summary of the chapter

- Excretory system (urinary system) is having two kidneys, two ureters, one urinary bladder and one urethra, each having typical function in the elimination of the metabolites formed in the body to the exterior.
- The clinical manifestations of urinary disease include one or more of the following: disturbances in urine volume (anuria, oliguria or polyuria); abnormalities of urinary sediments (RBCs, WBCs, casts and cryastals); abnormal excretion of serum proteins; reduction in glomerular filtration rate (azotemia); presence of hypertension and/or expanded total body volume (oedema); electrolyte abnormalities and pain or fever.
- In this chapter, we have discussed urinary tract infection; nephritic syndrome; nephrotic syndrome, acute kidney injury and chronic kidney disease in brief along with necessary physical examination performed in the patient and necessary investigations performed while doing the diagnosis of the diseases.

Self-Assessment: Questions for Reviews

Short essay questions

- Explain urinary tract infection
- Describe prostatomegaly
- Explain chronic kidney disease
- Explain nephritic syndrome
- Explain nephrotic syndrome

MCQs

1. The term polyuria is used when the 24 hours urine amounts more than
   a. 500 ml;
   b. 1,200 ml;
   c. 2,000 ml;
   d. 3,000 ml

Answer: (d)

2. When the urine volume is < 100 ml in 24 hours, the term used in this situation is
   a. Nocturia; b. Anuria;
   c. Oliguria; d. Polyuria

Answer: (b)

3. When an individual is presented with burning micturition, the probable organ afflicted will be
   a. Kidney; b. Ureter;
   c. Urinary bladder; d. Urethra

Answer: (d)

4. Renal osteodystrophy manifests in an individual because of
   a. Hyperthyroidism;
   b. Hyperparathyroidism;

c. Hyperpituitarism;

d. Hypergonadism

Answer: (b)

5. Haematuria is one of the essential feature of

a. Urinary tract infection;

b. Nephritic syndrome;

c. Nephrotic syndrome;

d. Chronic kidney disease

Answer: (a)

6. Which of the following symptom indicates the presence of benign prostatic hyperplasia?

a. Burning micturition;

b. Intermittency;

c. Haematuria;

d. Increased frequency

Answer: (b)

7. Which of the following symptom will not be present in the patient of benign prostatic hyperplasia?

a. Hesitancy;

b. Intermittency;

c. Terminal dribbling;

d. Painless micturition

Answer: (d)

8. PSA level is specifically useful in the assessment of

a. Bladder dysfunction;

b. Urethral dysfunction;

c. Prostatic dysfunction;

d. Ureteral dysfunction

Answer: (c)

9. Nocturia is absent in which of the following condition?

a. Diabetes mellitus;

b. Congestive cardiac failure;

c. Benign prostatic hyperplasia;

d. Acute diarrhoea

Answer: (d)

10. Periorbital oedema in the early morning is observed in which of the following condition?

a. Cardiac oedema;

b. Renal oedema;

c. Hepatic oedema;

d. Nutrition deficiency induced oedema

Answer: (b)

Chapter resources

- https://www.news-medical.net/health/Urinary-Tract-Infections-in-Children.aspx
- https://livayur.com/health-condition/prostatomegaly-symptoms-stages-causes-diagnosis-treatment/
- https://www.netmeds.com/health-library/post/nephrotic-syndrome-causes-symptoms-and-treatment
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- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
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